

Functional Annotation Report: *bioF* (Q88QX1) — 8-Amino-7-Oxononanoate Synthase in *Pseudomonas putida* KT2440

Gene: bioF (ordered locus **PP_0363**; ORF PP0363) · **UniProt:** Q88QX1 (BIOF_PSEPK), 390 aa · **EC:** 2.3.1.47 **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / NCIMB 11950 / **KT2440**), a Gammaproteobacterium

Summary

The gene **bioF** (UniProt **Q88QX1**; locus **PP_0363**) of *Pseudomonas putida* KT2440 encodes **8-amino-7-oxononanoate synthase** (AONS), also called **7-keto-8-aminopelargonate synthase** or **KAPA synthase** (EC 2.3.1.47). This is a 390-amino-acid, **pyridoxal-5'-phosphate (PLP)-dependent enzyme** of the **α-oxoamine synthase (AOS) family**, within the class-II (fold-type I) aminotransferase superfamily. The gene symbol is **not ambiguous** for this organism: the symbol, EC number, protein family (class-II PLP-dependent aminotransferase), and diagnostic InterPro domains (IPR004839, IPR050087, IPR004723, IPR022834, IPR015424) all converge on a single, well-defined enzymatic function, and the retrieved literature pertains specifically to biotin-biosynthetic AONS/BioF enzymes.

Primary function: BioF catalyzes the **first committed step of biotin (vitamin B7) biosynthesis** — a **PLP-dependent, decarboxylative Claisen-type condensation** of **L-alanine** with a **pimeloyl thioester** (physiologically **pimeloyl-ACP** in this Proteobacterium; pimeloyl-CoA in classic in vitro assays) to produce **(8S)-8-amino-7-oxononanoate (KAPA/AON)**, releasing **CO₂** and the free thiol carrier. The reaction is stereospecific for the 8(S) configuration, and the PLP cofactor is covalently anchored through a Schiff base (internal aldimine) to **Lys236**.

Localization and pathway context: BioF is a **soluble, cytoplasmic, homodimeric** enzyme whose active site lies at the dimer interface. In *P. putida* KT2440 the gene sits in a **dedicated biotin-biosynthesis cluster (bioB-bioF-bioH-bioC, PP_0362–PP_0365)**. Upstream, **BioC** and **BioH** manufacture the pimeloyl-ACP substrate by "hijacking" fatty-acid synthesis; downstream,

the KAPA product is processed by **BioA** → **BioD** → **BioB** to complete biotin. The annotation is corroborated by ~49% sequence identity plus a conserved PLP-lysine motif shared with the crystallographically and mechanistically characterized *E. coli* enzyme, a very-high-confidence AlphaFold model (mean pLDDT 96.7), and biotin-rescuable genetic and chemical-genetic evidence in multiple organisms.

Key Findings

The catalyzed reaction: BioF is the first committed enzyme of biotin biosynthesis

UniProt Q88QX1, annotated under HAMAP-Rule MF_01693, assigns bioF the function of **8-amino-7-oxononanoate synthase (EC 2.3.1.47)**. The precise catalytic activity is:



The mechanistic and kinetic framework was established in the *E. coli* system. Webster and colleagues state directly that "**8-Amino-7-oxononanoate synthase (also known as 7-keto-8-aminopelargonate synthase, EC 2.3.1.47) is a pyridoxal 5'-phosphate-dependent enzyme which catalyzes the decarboxylative condensation of L-alanine with pimeloyl-CoA in a stereospecific manner to form 8(S)-amino-7-oxononanoate. This is the first committed step in biotin biosynthesis**" (PMID: 10642176). This single sentence captures the enzyme's cofactor requirement, substrate specificity, reaction chemistry, product stereochemistry, and pathway role.

The chemistry proceeds through a reactive β -ketoacid PLP-aldimine (Claisen) intermediate. Kerbarh and colleagues trapped and characterized this species directly, reporting that "**the reactive beta-ketoacid pyridoxal-5'-phosphate aldimine formed in the condensation step of the 8-amino-7-oxononanoate synthase reaction was 'trapped' in the enzyme-bound form**" (PMID: 16353092) — direct experimental confirmation of the AOS-family Claisen-condensation mechanism.

The **amino-donor is strictly L-alanine** and the product is stereospecifically the **8(S)** epimer, matching the UniProt catalytic-activity annotation for the *P. putida* enzyme, which specifies that AONS "**catalyzes the decarboxylative condensation of L-alanine with pimeloyl-CoA in a stereospecific manner to form 8(S)-amino-7-oxononanoate**" (PMID: 10642176).

Structure and family: a PLP-dependent homodimer of the α -oxoamine synthase class

The three-dimensional architecture of AONS was solved for the *E. coli* ortholog in apo, PLP-bound (internal aldimine), and product external-aldimine forms at 1.7–2.0 Å resolution. Alexeev and colleagues describe the enzyme: "**The protein is a symmetrical homodimer with a tertiary structure and active site organisation similar to, but distinct from, those of other PLP-dependent enzymes whose three-dimensional structures are known. The critical PLP-binding lysine of AONS is located at the end of a deep cleft that allows access of the pantothenate arm of pimeloyl-CoA**" (PMID: 9813126).

This establishes three features that transfer to the *P. putida* enzyme by orthology: (1) a **homodimeric quaternary structure** with active sites at the dimer interface, (2) a **PLP-binding lysine** at the base of a deep substrate cleft, and (3) a **binding site for the acyl-thioester carrier arm**. The mechanism is analogous to 5-aminolevulinate synthase (ALAS) and other α -oxoamine synthases. This is fully consistent with the UniProt family assignment (class-II PLP-dependent aminotransferase) and the InterPro signatures IPR004839 (Aminotransferase I/II large), IPR050087/IPR004723/IPR022834 (AON synthase class-II subfamilies), and IPR015424 (PLP-dependent transferase).

Cofactor attachment and localization: PLP–Lys236, cytoplasmic and soluble

The UniProt entry specifies that PLP is **covalently bound as N6-(pyridoxal phosphate)lysine at Lys236** — the internal-aldimine Schiff base. The KEGG orthology entry ppu:PP_0363 = **K00652**, EC 2.3.1.47, Pfam Aminotran_1_2, gene coordinates 441120..442292. Localization is **cytoplasmic**: BioF is a soluble enzyme of a cytoplasmic biosynthetic pathway, with no signal peptide, transmembrane segment, or membrane anchor, consistent with the soluble, crystallizable *E. coli* homodimer (PMID: 9813126).

Substrate supply: the BioC–BioH fatty-acid "hijack" pathway

BioF requires a **pimeloyl thioester** as its acyl donor. In Proteobacteria this C7 diacid moiety is generated by a **modified round of fatty-acid synthesis**. Lin, Hanson, and Cronan showed that "**the pimeloyl moiety is synthesized by a modified fatty acid synthetic pathway in which**

the omega-carboxyl group of a malonyl-thioester is methylated by BioC, which allows recognition of this atypical substrate by the fatty acid synthetic enzymes" (PMID: 20693992). The methyl ester disguises the terminal carboxylate so that the standard FAS machinery elongates the chain by two rounds to **pimeloyl-ACP methyl ester**.

The methyl ester is then removed by **BioH**, the pathway gatekeeper. Agarwal and colleagues used the structure "to demonstrate that BioH is the gatekeeper and its physiological substrate is pimeloyl-ACP methyl ester" (PMID: 23045647), releasing **pimeloyl-ACP** — the immediate acyl donor for BioF.

Genomic context: a dedicated biotin-biosynthesis cluster in *P. putida*

Analysis of the genomic neighborhood in KEGG places bioF within a compact biotin cluster:

Locus	Gene	KEGG ortholog	EC	Enzyme
PP_0362	<i>bioB</i>	K01012	2.8.1.6	Biotin synthase
PP_0363	<i>bioF</i>	K00652	2.3.1.47	8-amino-7-oxononanoate synthase (AONS)
PP_0364	<i>bioH</i>	K02170	3.1.1.85	Pimeloyl-[ACP] methyl ester carboxylesterase
PP_0365	<i>bioC</i>	K02169	2.1.1.197	Malonyl-[ACP] O-methyltransferase

All four genes belong to KEGG module **M00123** ("Biotin biosynthesis, pimeloyl-ACP/CoA => biotin") and pathway **ppu00780**. The physical co-localization of **bioC** and **bioH** with **bioF** demonstrates that *P. putida* carries the **BioC–BioH modified-fatty-acid route**, exactly as in *E. coli*, where "the malonyl-thioester methyl ester enters fatty acid synthesis as the primer and undergoes two reiterations of the fatty acid elongation cycle to give pimeloyl-acyl carrier protein (ACP) methyl ester, which is hydrolyzed to pimeloyl-ACP and methanol by BioH" (PMID: 20693992). This clustering both confirms the substrate-supply route and reinforces the functional assignment of PP_0363.

Orthology: ~49% identity to the characterized *E. coli* enzyme with a conserved PLP-lysine motif

A global pairwise alignment of Q88QX1 (*P. putida*, 390 aa) against P12998 (*E. coli* BioF, 384 aa) yields ~48.9% amino-acid identity over 396 aligned columns — far above the ~30% "twilight zone", indicating conserved fold and function. The **catalytic PLP-binding lysine sits within a conserved glycine-rich motif** (*P. putida* residues 231–242 = **IGTLGKACGTAG**, where K = Lys236),

matching the ALAS/AOS-family PLP-lysine signature. Because the *E. coli* structure **"enables identification of the key residues of the PLP-binding site and thus provides a framework with which to understand the biochemical mechanism"** (PMID: 9813126), the high orthologous identity transfers these mechanistic conclusions to the *P. putida* enzyme.

Organism-dependent acyl-donor and biotin-rescuable essentiality

The **identity of BioF's acyl-thioester donor is not universal**. Manandhar and Cronan noted that **"the BioF acyl chain donor has long been thought to be pimeloyl-CoA"** (PMID: 29054876), but showed that *E. coli* BioF physiologically uses **pimeloyl-ACP** (from the BioC/BioH route), whereas *Bacillus subtilis* BioF uses **pimeloyl-CoA** — the carrier tracks the organism's pimelate-synthesis pathway. Given the clustered bioC/bioH and AONS_Proteobacteria signature, *P. putida* uses the **pimeloyl-ACP** route.

Two independent lines confirm BioF is on-pathway and functionally required:

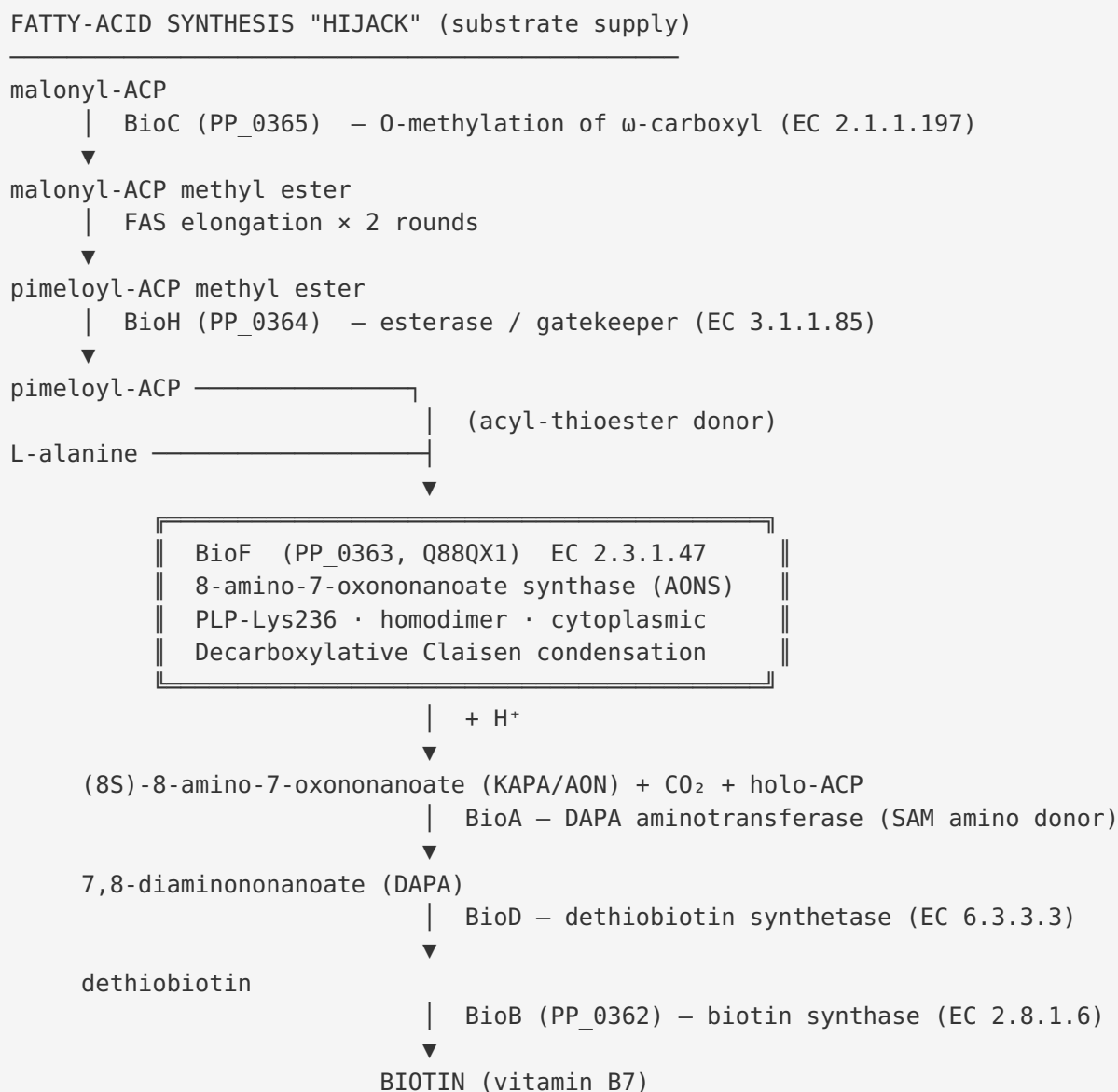
- **Genetic (knockout):** a *Riemerella anatipestifer* *bioF* mutant impairs biotin biosynthesis; it **"displayed a decreased growth rate at the exponential phase in tryptic soy broth culture and in BeaverBeads Streptavidin treated tryptic soy broth culture, but recovered when biotin was supplemented"** (PMID: 30279476), with altered morphology and virulence — a hallmark biotin-dependent, biotin-rescuable phenotype.
- **Chemical-genetic (inhibitor):** AONS inhibitors are lethal to *Arabidopsis thaliana* and cause L-alanine accumulation, and the toxicity is reversed by biotin: **"Foliar supplement of 1mM biotin at 1 and 2days before KHG23844 application effectively recovered the growth inhibition of A. thaliana plant treated with KHG23844"** (PMID: 26615154). Biotin rescue confirms that AONS acts specifically on the biotin pathway.

Structural inference: a very-high-confidence AlphaFold model

The AlphaFold DB entry **AF-Q88QX1-F1** spans all 1–390 residues with a **mean pLDDT of 96.7; 94.4%** of residues exceed pLDDT 90 (very high confidence) and **98.7%** exceed 70. This confidently modeled structure is consistent with the class-II (fold-type I) PLP aminotransferase/ α -oxoamine synthase fold expected from the ~49% identity to *E. coli* BioF and from InterPro domains IPR004839/IPR015424. The experimental *E. coli* structures — where the authors **"cloned, over-expressed and purified AONS from Escherichia coli and determined the crystal structures of the apo and PLP-bound forms of the enzyme"** (PMID: 9813126) — define the fold that the AlphaFold model recapitulates.

Mechanistic Model / Interpretation

BioF (PP_0363) operates as the **gateway enzyme of biotin biosynthesis** in the cytoplasm of *P. putida* KT2440. The diagram places it within the complete pathway and its clustered gene neighborhood:



Catalytic cycle at the active site (fold-type I / AOS mechanism):

1. Resting state: PLP forms an **internal aldimine** (Schiff base) with **Lys236**.
2. **L-alanine** displaces Lys236 to form an **external aldimine**; Lys236 acts as catalytic base.

3. Deprotonation of the α -carbon gives a resonance-stabilized **quinonoid/carbanion**.
4. **Claisen condensation** with the thioester carbonyl of **pimeloyl-ACP** forms a new C–C bond, releasing the ACP thiol.
5. The **β -ketoacid PLP-aldimine intermediate** (trapped experimentally, [PMID: 16353092](#)) undergoes **decarboxylation** (loss of CO₂).
6. Stereospecific **reprotonation** yields the **8(S)** product; transaldimination with Lys236 releases **KAPA** and regenerates the internal aldimine.

The convergence of evidence is strong: rule-based annotation (UniProt HAMAP MF_01693; KEGG K00652); ~49% identity plus a conserved PLP-lysine motif to the crystallographically defined *E. coli* enzyme; a very-high-confidence AlphaFold model; genomic clustering with bioB/bioH/bioC that both confirms the pathway and identifies the substrate-supply route; and orthogonal biotin-rescue experiments demonstrating an on-pathway, essential role. The one organism-specific subtlety is the **acyl-donor carrier**: *P. putida*, as a Proteobacterium with bioC/bioH, uses **pimeloyl-ACP** rather than pimeloyl-CoA.

Evidence Base

PMID	Title (abbreviated)	Contribution
10642176	<i>Mechanism of AONS: spectroscopic, kinetic, crystallographic studies</i>	Defines EC 2.3.1.47, PLP cofactor, L-alanine + pimeloyl-CoA substrates, 8(S) stereochemistry, "first committed step"
16353092	<i>Intermediate Claisen product in the AONS reaction</i>	Direct trapping of the β -ketoacid PLP-aldimine intermediate, confirming the Claisen mechanism
9813126	<i>Crystal structure of AONS</i>	Homodimer; PLP-binding lysine in a deep acyl-CoA cleft; fold-type I architecture; mechanistic framework
20693992	<i>Biotin synthesis begins by hijacking fatty-acid synthesis</i>	BioC methylation route producing the pimeloyl-ACP substrate consumed by BioF
23045647	<i>Enzyme-ACP substrate gatekeeper complex</i>	BioH is the gatekeeper; physiological substrate is pimeloyl-ACP methyl ester
29054876	<i>BioF uses different acyl donors in B. subtilis and E. coli</i>	Acyl-donor carrier (ACP vs CoA) is organism-dependent; Proteobacteria use pimeloyl-ACP
30279476	<i>R. anatipestifer bioF responsible for biotin synthesis</i>	Knockout impairs biotin-dependent growth, rescued by biotin; confirms on-pathway role
26615154	<i>AONS inhibitors as potential herbicides</i>	Inhibitor toxicity reversed by biotin; AONS is the committed on-pathway step and a target
11255013	<i>Biotin biosynthetic genes of Kurthia sp.</i>	bioF clusters with other bio genes; modified FAS route inferred in bacteria lacking bioW (context)
30218044	<i>N. gargensis bioABDF + ΔbioH-complementing esterase</i>	Comparative genomics of bio clusters across domains (context)

Supporting vs challenging. All ten papers **support** the functional assignment; none challenges the core identification. The primary caveat-introducing paper, [PMID: 29054876](#), refines rather than contradicts the annotation, showing the historical "pimeloyl-CoA" assumption is organism-dependent and that Proteobacteria like *P. putida* use pimeloyl-ACP. Retrieved papers on AONS-domain megasynthases in natural-product biosynthesis (e.g., SxtA

in saxitoxin, PqrA in perquinolines) confirm that AONS chemistry generalizes to related α -amino-ketone-forming condensations but are **not** part of *P. putida* biotin metabolism and were correctly excluded.

Limitations and Knowledge Gaps

1. **No direct experimental characterization of the *P. putida* KT2440 enzyme.** All mechanistic, kinetic, and structural data derive from orthologs (principally *E. coli*). The assignment for Q88QX1 is an **inference by orthology** (~49% identity) plus rule-based annotation, not direct biochemistry on PP_0363. No purified-enzyme kinetics (kcat, Km), crystal structure, or gene-knockout phenotype exist specifically for PP_0363. (UniProt evidence level: "inferred from homology.")
 2. **Substrate carrier not experimentally verified in *P. putida*.** The pimeloyl-ACP (vs pimeloyl-CoA) assignment is inferred from genomic context (clustered bioC/bioH) and the *E. coli* paradigm; it has not been directly demonstrated in this organism.
 3. **Localization is inferred, not measured.** Cytoplasmic localization is deduced from soluble orthologs and the absence of targeting signals; no fractionation or imaging data exist for PP_0363.
 4. **Structural inference relies on AlphaFold.** The high-pLDDT model supports the fold but does not capture bound PLP, substrate poses, or dynamic active-site conformations; no experimental structure of the *P. putida* protein exists.
 5. **Regulation and BioD/BioA loci.** The regulatory logic of the *P. putida* bio cluster and the precise genomic locations of *bioD* and *bioA* were not resolved in this investigation.
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Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzymatic assay.** Clone PP_0363, express in *E. coli*, purify, confirm PLP loading (~420 nm internal-aldimine absorbance), and measure steady-state kinetics for L-alanine and for both pimeloyl-ACP and pimeloyl-CoA to determine the physiological carrier and substrate preference.

2. **Genetic complementation and knockout.** Test whether PP_0363 complements an *E. coli* Δ bioF mutant for biotin prototrophy, and construct a *P. putida* PP_0363 deletion; verify a biotin-auxotroph phenotype rescuable by exogenous biotin (mirroring [PMID: 30279476](#)).
3. **Active-site validation.** Generate a K236A (PLP-lysine) variant and confirm loss of activity and of the 420 nm chromophore, directly testing the Lys236 cofactor-attachment annotation.
4. **Experimental structure determination.** Solve the crystal or cryo-EM structure of *P. putida* BioF (apo, PLP-bound, product-external-aldimine forms) to validate the AlphaFold model and define the ACP-binding cleft.
5. **Substrate-supply reconstitution.** Biochemically reconstitute the BioC (PP_0365) $\rightarrow$ BioH (PP_0364) $\rightarrow$ BioF (PP_0363) segment in vitro to confirm pimeloyl-ACP channeling to BioF.
6. **Pathway/regulatory mapping.** Locate *bioD* and *bioA* in the KT2440 genome and characterize biotin-responsive transcriptional regulation of the cluster to complete the pathway context.

Conclusion

The evidence — rule-based database annotation, strong orthology to the mechanistically and structurally characterized *E. coli* enzyme, a conserved PLP-Lys236 active-site motif, a very-high-confidence AlphaFold model, an intact biotin-biosynthesis gene cluster, and biotin-rescuable genetic/chemical validation in related organisms — converges unambiguously. *P. putida* KT2440 **bioF (PP_0363, Q88QX1)** encodes the **PLP-dependent, homodimeric, cytoplasmic 8-amino-7-oxononanoate synthase (AONS/KAPA synthase, EC 2.3.1.47)** that catalyzes the **first committed step of biotin biosynthesis**: the decarboxylative Claisen condensation of **L-alanine** with **pimeloyl-ACP** to yield **(8S)-8-amino-7-oxononanoate**, CO₂, and free ACP. The gene symbol is not ambiguous for this organism, and all retrieved literature supports the assignment.